Talk2MeLatin

Although Latin is no longer used in daily conversation, it continues to be widely studied by many students and scholars today. Those learning Latin face a unique challenge, which is that unlike modern languages, there are no native speakers to practise with or easily ask questions when they encounter grammatical and syntactical issues in Latin texts. Available digital tools typically only offer dictionary-style lookups. Artificial Intelligence (AI) has become a familiar part of modern education, especially in language learning. Many students today benefit from interactive tools that provide instant feedback and support. However, this progress has largely focused on widely spoken modern languages, leaving classical languages such as Latin behind. This project aims to contribute to this gap by creating a chatbot system where a student can ask a question in English about a Latin word and receive an accurate and helpful answer about its grammatical and syntactical features.

The system first needs to understand what the student is asking in order to produce a suitable answer. For example, a learner might ask whether a word is a noun or a verb, or what tense a verb is in. To do this, it uses an intent classifier, which determines what kind of grammatical question the student is asking. A question like "What is the tense of *amavit*?" would be classified as a tense question, while "Is *puella* singular or plural?" would be classified as a number question. This classifier was built by fine-tuning a BERT language model on a custom dataset created within the project to specifically understand possible grammar questions.

In addition to this, a morphological analyser was also implemented. This is a model that takes a Latin word and predicts its set of grammatical features, such as its part of speech, case, number, gender, tense, mood, and voice. This model was trained on a dataset of annotated sentences from Latin texts, and was built by using a specialised version of BERT trained specifically on Latin.

A key part of the project involved building the data needed to support the system. The UniMorph dataset was used as it provides a large collection of morphologically annotated Latin word forms. This dataset was then further extended to include additional morphological features to create a more comprehensive dataset. Additionally, a separate dataset of grammar-related student questions was generated to train the intent classification model to recognise different ways students might phrase their questions. This was particularly important because learners often ask the same question in many different ways.

Finally, once the question is understood and the system retrieves the relevant grammatical information about the Latin word, an answer is constructed and returned by the chatbot. A conversational user interface was also developed where students can type questions and receive responses in a chat format.

Overall, this project demonstrates how AI can be applied to support the learning of Latin. An interactive tool has the potential to make Latin grammar more approachable and engaging, helping make this classical language more accessible to students.

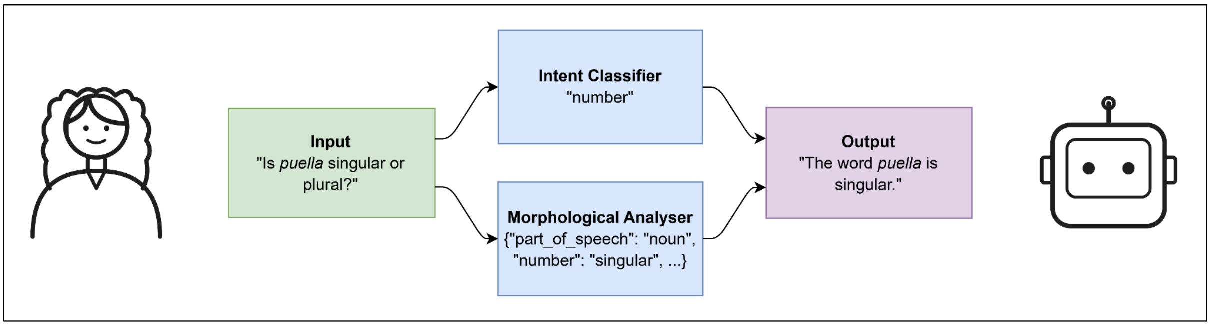

Figure 1: Overview of the system workflow.